nearly the same as the bottom (inner) red orbit in Figures 6.3 and 6.4. The next orbit in Figure 6.5 winds around a slightly larger sphere, in a nearly polar and slightly forward direction. The third orbit is larger still, but backward and nearly polar. The fourth is very nearly equatorial and backward, that is, nearly the upper (outer) red equatorial orbit of Figures 6.3 and 6.4. These orbits are actually inside each other; I pulled them apart so they are easier to see.

Some photons that are temporarily trapped in the shell of fire escape outward; they spiral away from Gargantua. The rest escape spiraling inward; they spiral toward Gargantua and plunge through its horizon. The nearly trapped but escaping photons have a big impact on Gargantua's visual appearance in *Interstellar*. They mark the edge of Gargantua's shadow as seen by the *Endurance*'s crew, and they produce a thin bright line along the shadow's edge: a "ring of fire" (Chapter 8).

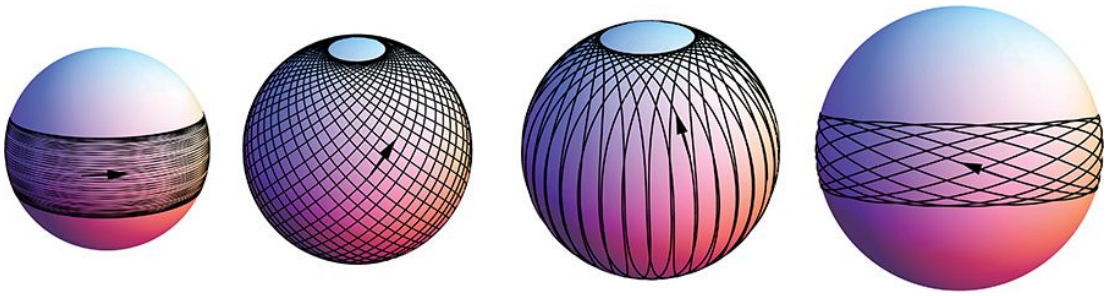


Fig. 6.5. Examples of light rays (photon orbits) temporarily trapped in the shell of fire, as computed using Einstein's relativistic equations.

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13 The literal French translation of "a black hole has no hair" is so obscene that French publishers resisted it vigorously, to no avail.

14 For some quantitative details, see *Some Technical Notes*, at the end of this book.

15 A more reasonable value might be 200 million times the Sun's mass, but I want to keep the numbers simple and there's a lot of slop in this one, so I chose 100 million.